

## Lab 4 plus

**NAME:** LIM YU HAN

**MATRIC NO:** A23CS0241

\*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

## Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

### Objectives

- To explore ARP and switching operations.

### Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

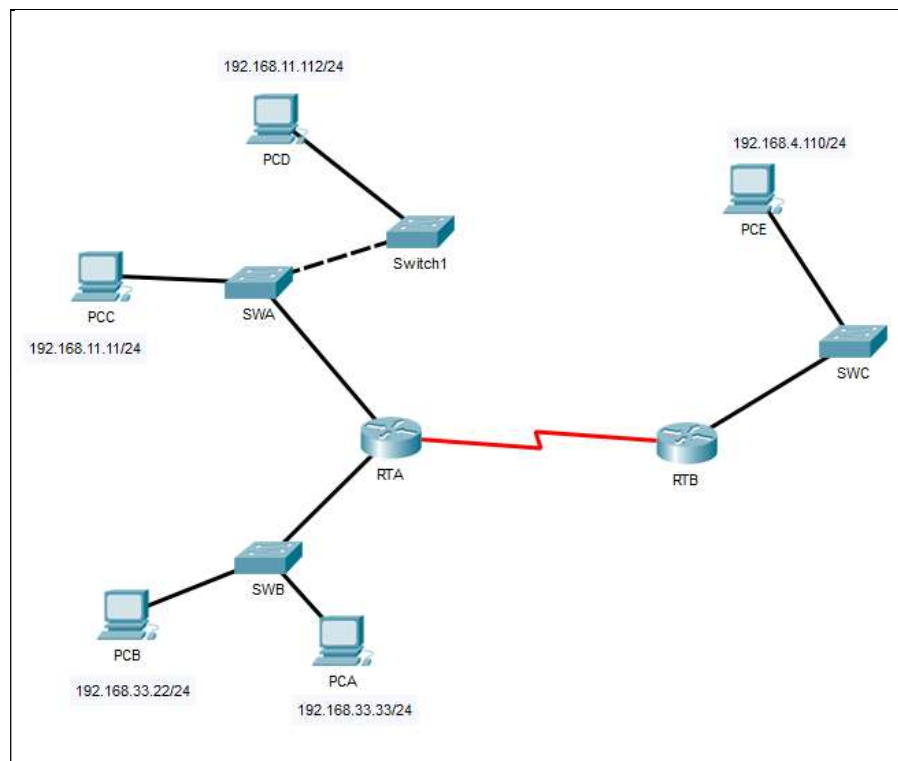


Figure 1

## Part 1: Review the topology

**Step 1:** Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

| Protocol | Address      | Age (min) | Hardware Addr  | Type | Interface       |
|----------|--------------|-----------|----------------|------|-----------------|
| Internet | 192.168.11.1 | -         | 0002.4A00.0E91 | ARPA | FastEthernet1/0 |
| Internet | 192.168.33.1 | -         | 000C.CF0C.593A | ARPA | FastEthernet0/0 |

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

| Protocol | Address     | Age (min) | Hardware Addr  | Type | Interface       |
|----------|-------------|-----------|----------------|------|-----------------|
| Internet | 192.168.4.1 | -         | 0001.977A.B614 | ARPA | FastEthernet0/0 |

- c. At Switches SWA, SWB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

### SWA

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0002.4a00.0e91 | DYNAMIC | Fa0/1 |
| 1                 | 000c.8546.7d85 | DYNAMIC | Fa1/1 |

### SWB

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 000c.cf0c.593a | DYNAMIC | Fa0/1 |

### SWC

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0001.977a.b614 | DYNAMIC | Fa0/1 |

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
No ARP Entries Found
```

PCB

```
No ARP Entries Found
```

PCC

```
No ARP Entries Found
```

PCD

```
No ARP Entries Found
```

PCE

```
No ARP Entries Found
```

- e. What are your thoughts on the results?

When running `arp -a` on PCA, PCB, PCC, PCD, and PCE, the ARP table displays the mapping of IP addresses to MAC addresses for devices the PC has communicated with. Each ARP table shows dynamic entries created during recent communications. If an IP address is not in the table, it indicates no communication has occurred with that device yet. No entry for all PCs because all the PCs didn't receive any ARP request. There will be no entry in the PC ARP table.

## Part 2: Generate Network Traffic

### Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=8ms TTL=128
Reply from 192.168.33.22: bytes=32 time=4ms TTL=128
Reply from 192.168.33.22: bytes=32 time=4ms TTL=128
Reply from 192.168.33.22: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms
```

- c. In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
```

- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic
```

- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

## Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time=8ms TTL=127
Reply from 192.168.33.22: bytes=32 time=8ms TTL=127
Reply from 192.168.33.22: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic

```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time=12ms TTL=128
Reply from 192.168.11.112: bytes=32 time=6ms TTL=128
Reply from 192.168.11.112: bytes=32 time=6ms TTL=128
Reply from 192.168.11.112: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 12ms, Average = 7ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic

```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 10ms, Average = 10ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic

```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands.  
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

| Protocol | Address       | Age (min) | Hardware Addr  | Type | Interface       |
|----------|---------------|-----------|----------------|------|-----------------|
| Internet | 192.168.11.1  | -         | 0002.4A00.0E91 | ARPA | FastEthernet1/0 |
| Internet | 192.168.11.11 | 0         | 00D0.D39A.C0D9 | ARPA | FastEthernet1/0 |
| Internet | 192.168.33.1  | -         | 000C.CF0C.593A | ARPA | FastEthernet0/0 |
| Internet | 192.168.33.22 | 0         | 0060.47EA.A746 | ARPA | FastEthernet0/0 |

- g. At Router RTA, enter the CLI. At the command prompt type the following commands.  
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

| Protocol | Address       | Age (min) | Hardware Addr  | Type | Interface       |
|----------|---------------|-----------|----------------|------|-----------------|
| Internet | 192.168.4.1   | -         | 0001.977A.B614 | ARPA | FastEthernet0/0 |
| Internet | 192.168.4.110 | 0         | 0060.702D.7C08 | ARPA | FastEthernet0/0 |

### Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.  
Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0001.6462.0278 | DYNAMIC | Fa1/1 |
| 1                 | 0002.4a00.0e91 | DYNAMIC | Fa0/1 |
| 1                 | 000c.8546.7d85 | DYNAMIC | Fa1/1 |
| 1                 | 00d0.d39a.c0d9 | DYNAMIC | Fa2/1 |

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0002.1755.9a06 | DYNAMIC | Fa2/1 |
| 1                 | 000c.cf0c.593a | DYNAMIC | Fa0/1 |
| 1                 | 0060.47ea.a746 | DYNAMIC | Fa1/1 |

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0001.977a.b614 | DYNAMIC | Fa0/1 |
| 1                 | 0060.702d.7c08 | DYNAMIC | Fa1/1 |

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
```

| Mac Address Table |                |         |       |
|-------------------|----------------|---------|-------|
| Vlan              | Mac Address    | Type    | Ports |
| 1                 | 0005.5e95.187b | DYNAMIC | Fa0/1 |

- d. Do switches use arp table? (Y/N)

Yes

- e. Explain your answer in (d) *\*Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches utilize an ARP table to store the IP addresses and corresponding MAC addresses of network devices. This table serves the purpose of identifying the destination MAC addresses for communication with network nodes.

- f. What information does the command `show mac-address-table` gives?

To view MAC address table entries on a switch, the term "DYNAMIC" specifically pertains to displaying MAC addresses that have been learned by the switch during its operation.

### Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

#### Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. *\*Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a central switch (S1) connected to three PCs (PC1, PC2, PC3) and a wireless router (WRS2). PC1 is in VLAN 10 (172.17.10.21/24), PC2 is in VLAN 20 (172.17.20.22/24), and PC3 is in VLAN 88 (172.17.40.100/24). WRS2 is in VLAN 88 (172.17.88.25/24). A subinterface table for S1 is also visible.

On the right, the 'Packet Tracer - Configuring Wireless LAN Access' activity wizard is open. It shows an 'Addressing Table' with the following data:

| Device | Interface | IP Address   | Subnet Mask   | Default Gate |
|--------|-----------|--------------|---------------|--------------|
| R1     | G0/0.10   | 172.17.10.1  | 255.255.255.0 | N/A          |
|        | G0/0.20   | 172.17.20.1  | 255.255.255.0 | N/A          |
|        | G0/0.88   | 172.17.88.1  | 255.255.255.0 | N/A          |
| PC1    | NIC       | 172.17.10.21 | 255.255.255.0 | 172.17.10.1  |

At the bottom right of the activity wizard, a red box highlights the completion status: 'Completion: 100/100' and '1/1'.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.



